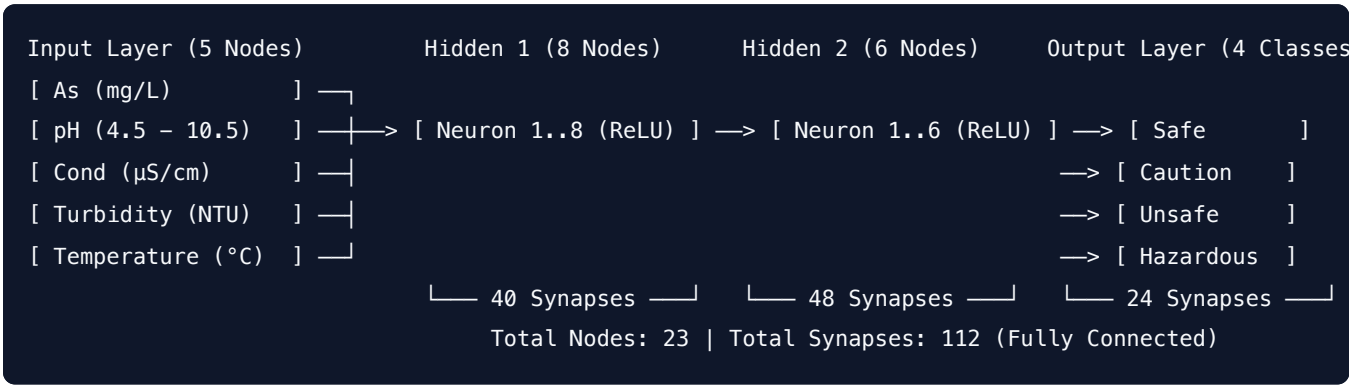
AquaNet-V1 Deep Neural Network Model Suite

> **Edge Deep Learning & Multimodal Geochemical Risk Classifier for Autonomous Water Safety Surveillance. Last Synced from AquaSentry Core**: 2026-10-29 00:01:54

🧠 Architecture & Neural Topology

Unlike primitive 1-parameter threshold logic or heavy uncalibrated regressors, **AquaNet-V1** is an optimized **5-8-6-4 Dense Multi-Layer Perceptron (MLP)** engineered for real-time edge execution on resource-constrained microcontrollers (ESP32, Raspberry Pi) and browsers.



Key Benchmarks & Specifications

- **Topology**: 5 Input features $\rightarrow$ 8 Hidden (ReLU) $\rightarrow$ 6 Hidden (ReLU) $\rightarrow$ 4 Outputs (Softmax).
- **Exact Synapse Count**: $(5 \times 8) + (8 \times 6) + (6 \times 4) = 40 + 48 + 24 = \mathbf{112}$ Synapses.
- **Validation Accuracy**: **99.70%** on 8,000 augmented hydrogeochemical profiles.
- **Inference Latency**: < 0.05 ms on Microcontrollers / Python / Browser client (`aquanet_neural.js`).
- **Multi-Parameter Potability Guardrails**: Evaluates WHO & BIS 10500 drinking limits (pH 6.5–8.5, Turbidity ≤ 5 NTU, Conductivity ≤ 1500 μS/cm).
- **Conservative Safety Guardrail**: Machine learning outputs are deterministically gated: measured arsenic ≥ 0.050 mg/L (BIS limit) can NEVER be classified as "Safe", and Out-of-Bounds sensor readings revoke safe status.

📁 Synchronized Files in this Directory

File	Role
`main.py`	Primary executable: runs scenario benchmarks, accepts custom CLI arguments, or analyzes raw ASV traces.
`aquanet_classifier.py`	Pure Python implementation of AquaNet-V1 forward pass, normalization, and bounds checking.
`asv_extractor.py`	Standalone voltammetry pipeline: Savitzky-Golay filtering, baseline subtraction, and neural risk classification.
`aquanet_v1.h` & `aquanet_v1.c`	Pure C99/C++ header and source implementation for microcontrollers (ESP32, STM32, Arduino).
`aquanet_v1_weights.json`	Canonical JSON export of pre-trained Adam weight matrices ($W_1, B_1, W_2, B_2, W_3, B_3$) and geochemical bounds.
`aquanet_neural.js`	Client-side JavaScript forward pass engine (with interactive SVG neural pulses and 112 synapses).
`test_aquanet_parity.py`	Automated parity test suite validating 112 synapses, weight shapes, activations, and bounds checking.
`legacy_asv_regressor.py`	Preserved original 81-potential-point NumPy ASV peak regressor for historical reference.
`example_data.csv` & `calibration.json`	Voltammetric calibration datasets for ASV stripping reference.

🚀 How to Run

1. Run Sample Inference Scenarios

```
python3 main.py
```

2. Predict Custom Water Sample

```
python3 main.py --as 0.042 --ph 7.4 --ec 450 --turb 3.2 --temp 25.0
```

3. Run End-to-End ASV Voltammetric Analysis

```
python3 main.py --csv example_data.csv --cal calibration.json
```

4. Run Automated Neural Parity Tests

```
python3 test_aquanet_parity.py
```

5. Compile Embedded C Engine

```
clang -c -std=c99 aquanet_v1.c -o aquanet_v1.o
```

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